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# CYCLISTIC DATA ANALYSIS PROJECT (R)
# Case Study: Google Data Analytics Professional Certificate
#
# Author: YAOVI SELOM GOUMENOU
# Role: Specialist in Business Intelligence | Data Analyst
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# -----
# 1. LIBRARIES
# -----
library(tidyverse) # Data manipulation and visualization
library(lubridate) # Date and time management

# -----
# 2. DATA EXPLORATION
# -----
# Overview of dataset structures
str(Divvy_Trips_2019_Q1)
str(Divvy_Trips_2020_Q1)

glimpse(Divvy_Trips_2019_Q1)
glimpse(Divvy_Trips_2020_Q1)

# Check dimensions
dim(Divvy_Trips_2019_Q1)
dim(Divvy_Trips_2020_Q1)

# Inspect column names
names(Divvy_Trips_2019_Q1)
names(Divvy_Trips_2020_Q1)

# -----
# 3. DATA CLEANING & STANDARDIZATION
# -----
# Cleaning 2019 data: renaming columns and selecting core variables
Divvy_2019 <- Divvy_Trips_2019_Q1 %>%
  rename(
    ride_id = trip_id,
    started_at = start_time,
    ended_at = end_time,
    member_casual = usertype,
    start_station_name = from_station_name,
    end_station_name = to_station_name
  ) %>%
  select(ride_id, started_at, ended_at, member_casual)

# Cleaning 2020 data: selecting matching variables
Divvy_2020 <- Divvy_Trips_2020_Q1 %>%
  select(ride_id, started_at, ended_at, member_casual)

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# 4. MERGE DATASETS
# -----
Divvy_total <- bind_rows(Divvy_2019, Divvy_2020)

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# 5. DATA CLEANING (FINAL DATASET)
# -----
Divvy_clean <- Divvy_total %>%
  # Remove rows with missing essential data
  filter(
    !is.na(started_at),
    !is.na(ended_at),
    !is.na(member_casual)
  ) %>%
  # Format dates and create new analytical columns
  mutate(
    started_at = ymd_hms(started_at),
    ended_at = ymd_hms(ended_at),
    duree_trajet_min = as.numeric(difftime(ended_at, started_at, units =
"mins")),
    jours_de_la_semaine = wday(started_at, label = TRUE, abbr = FALSE,
week_start = 1)
  ) %>%
  # Filter out trips with invalid durations (negative or > 24 hours)
  filter(
    duree_trajet_min > 0,
    duree_trajet_min <= 1440
  ) %>%
  # Standardizing user type labels
  mutate(
    member_casual = case_when(
      member_casual %in% c("Subscriber", "member") ~ "Subscriber",
      member_casual %in% c("Customer", "casual") ~ "Customer",
      TRUE ~ NA_character_
    )
  )

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# 6. EXPLORATORY ANALYSIS
# -----
# Verify unique user types
unique(Divvy_clean$member_casual)

# General duration statistics
summary(Divvy_clean$duree_trajet_min)

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# 7. AGGREGATED ANALYSIS
# -----
# Analysis by user type
moyenne_trajets <- Divvy_clean %>%
  group_by(member_casual) %>%
  summarise(
    duree_moyenne_min = mean(duree_trajet_min, na.rm = TRUE),
    duree_max_min = max(duree_trajet_min, na.rm = TRUE),
    duree_min_min = min(duree_trajet_min, na.rm = TRUE),
    nombre_trajets = n()
  )

# Analysis by day of the week
trajets_par_jour <- Divvy_clean %>%
  group_by(jours_de_la_semaine) %>%
  summarise(
    nombre_trajets = n(),
    duree_moyenne_min = mean(duree_trajet_min, na.rm = TRUE)
  ) %>%
  arrange(match(
    jours_de_la_semaine,
    c("lundi", "mardi", "mercredi", "jeudi", "vendredi", "samedi", "dimanche")
  ))

# -----
# 8. VISUALIZATIONS
# -----
# Chart 1: Total rides per day of the week
barplot(
  trajets_par_jour$nombre_trajets,
  names.arg = trajets_par_jour$jours_de_la_semaine,
  col = "lightcoral",
  main = "Total Rides per Day of the Week",
  ylab = "Number of Rides",
  las = 2,
  cex.axis = 0.8
)

# Chart 2: Average duration per user type
barplot(
  moyenne_trajets$duree_moyenne_min,
  names.arg = moyenne_trajets$member_casual,
  col = "lightgreen",
  main = "Average Trip Duration by User Type",
  ylab = "Duration (minutes)"
)

#
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# END OF SCRIPT
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